ALGEBRA 2 · UNIT 5: RADICALS & RATIONAL EXPONENTS

Crash Scene Investigator

Use radical formulas and rational exponents to reconstruct an accident and study motion.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

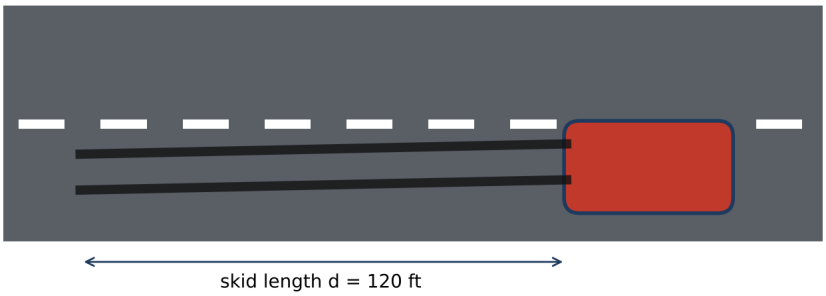
The Scenario

A car skidded 120 feet on a dry road before stopping in a 45-mph zone. Accident investigators use s = √(30df), where s is speed in mph, d is skid length in feet, and f is the road's friction factor (dry asphalt 0.75, wet asphalt 0.45).

Your lab will estimate speeds, solve for unknowns, and explore other formulas with radicals and rational exponents, then write a report.

Crash Lab Formula

speed $s = \sqrt{(30 \cdot d \cdot f)}$ (mph; d in feet; f = road friction factor)



Learning Goals

- I can evaluate and simplify radical expressions.
- I can solve radical equations and check for extraneous solutions.
- I can use radical formulas in science.

Project Timeline

When	What you will do
Day 1	The Skid Marks
Day 2	Solve & Check
Day 3	Pendulums & Report
Final	Investigation Report + reflection

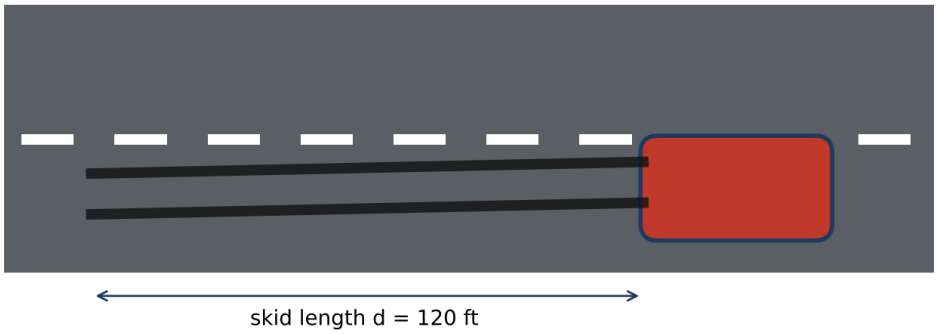
Part 1: The Skid Marks

Day 1 · Use s = √(30df).

1. Find the speed for d = 120 ft on dry asphalt (f = 0.75). Write the exact answer in simplest radical form and as a decimal.

Crash Lab Formula

speed s = √(30 · d · f) (mph; d in feet; f = road friction factor)



2. Was the driver speeding in the 45-mph zone?

3. Find the speed for the same skid on wet asphalt (f = 0.45). Simplify the radical.

4. How long would the skid be on dry asphalt at exactly 45 mph? Solve 45 = √(22.5d).

Part 2: Solve & Check

Day 2 · Square both sides, then check every answer.

5. Solve $\sqrt{(x + 7)} = x - 5$. Check both answers and identify any extraneous solution.

6. Explain why squaring both sides can create an answer that doesn't work.

7. Make a table for $s(d) = \sqrt{(22.5d)}$ with d = 0, 40, 90, 160 and sketch the graph.

Part 3: Pendulums & Report

Day 3 · A pendulum's period is $T = 2\pi\sqrt{(L \div 32)}$, with L in feet and T in seconds.

8. Find the period of an 8-ft pendulum.

9. Find the period of a 2-ft pendulum. How does making the pendulum 4 times shorter change the period?

10. Write a short report on the crash case using your answers.

Final Product: Investigation Report

Your final product must include:

- Speed calculations with simplified radicals
- Radical equation with check
- Graph of the speed function
- Case report

Reflection

R1. Why do we have to check answers to radical equations?

R2. Why does a 4-times-longer skid only double the speed?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Conclusions are supported by correct radical calculations, and any extraneous solutions are identified. Goes beyond the minimum with a creative, realistic choice.	Conclusions are supported by correct radical calculations, and any extraneous solutions are identified.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Polished, organized, and visually engaging.	Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.